Arduino MEGA 2560 With WiFi Built-in - ESP8266

By Fernando Koyanagi in Circuits > Arduino 250,072 81 97

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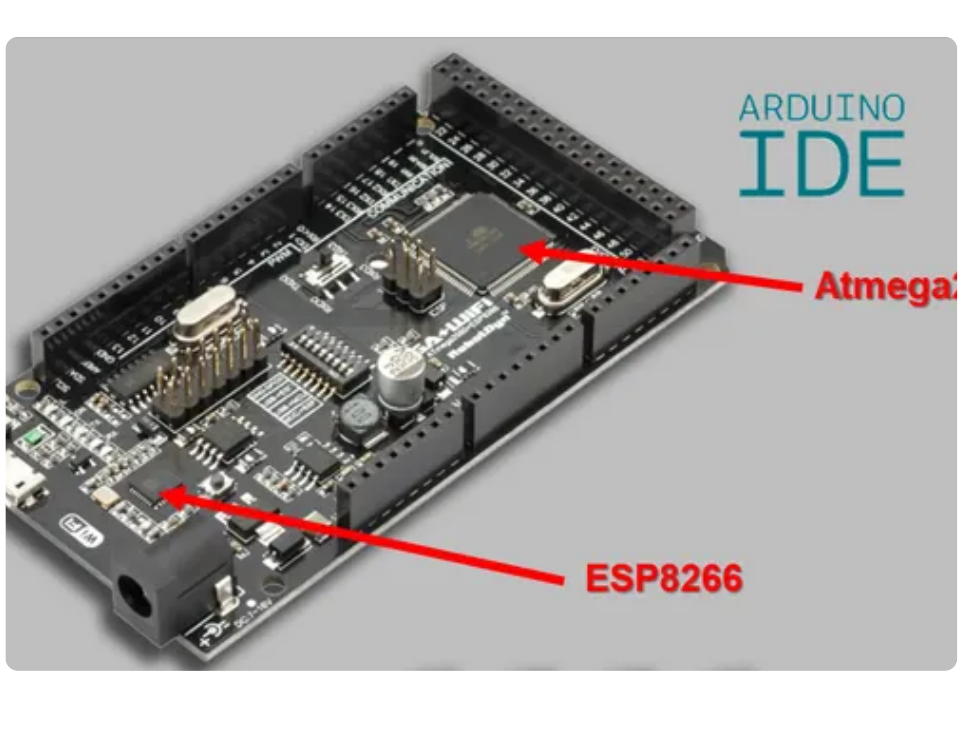
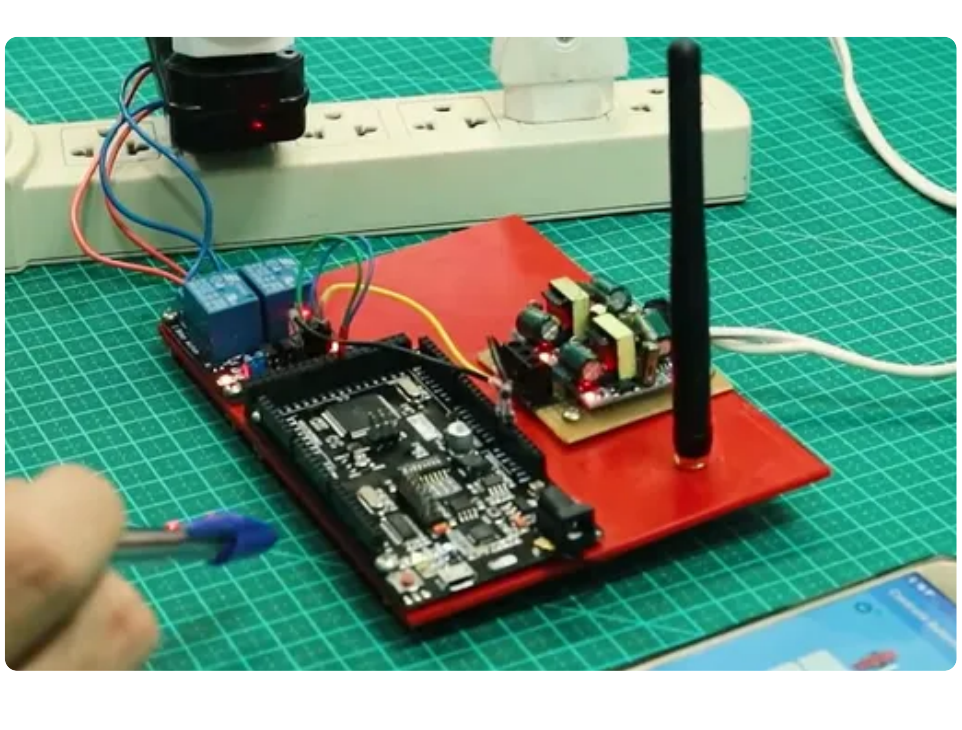

ARDUINO IDE
Atmega
ESP8266



By Fernando Koyanagi
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In today's text, we discuss an Arduino that I consider extremely special, as it has an ESP8266 embedded in its board. It doesn't have the ESP12 soldered onto the board. Instead, it has the Espressif chip. So, on the board you have the built-in Tensilica chip with 4MB of memory, along with the ATmega2560, which is the traditional Arduino Mega.

Let's move onto how this Arduino works, and let's do an assembly that shows when you should select ESP or Mega to perform a home automation. With this, we can turn lamps on and off, which is a mechanism that can be very useful for you to make improvements in your house.

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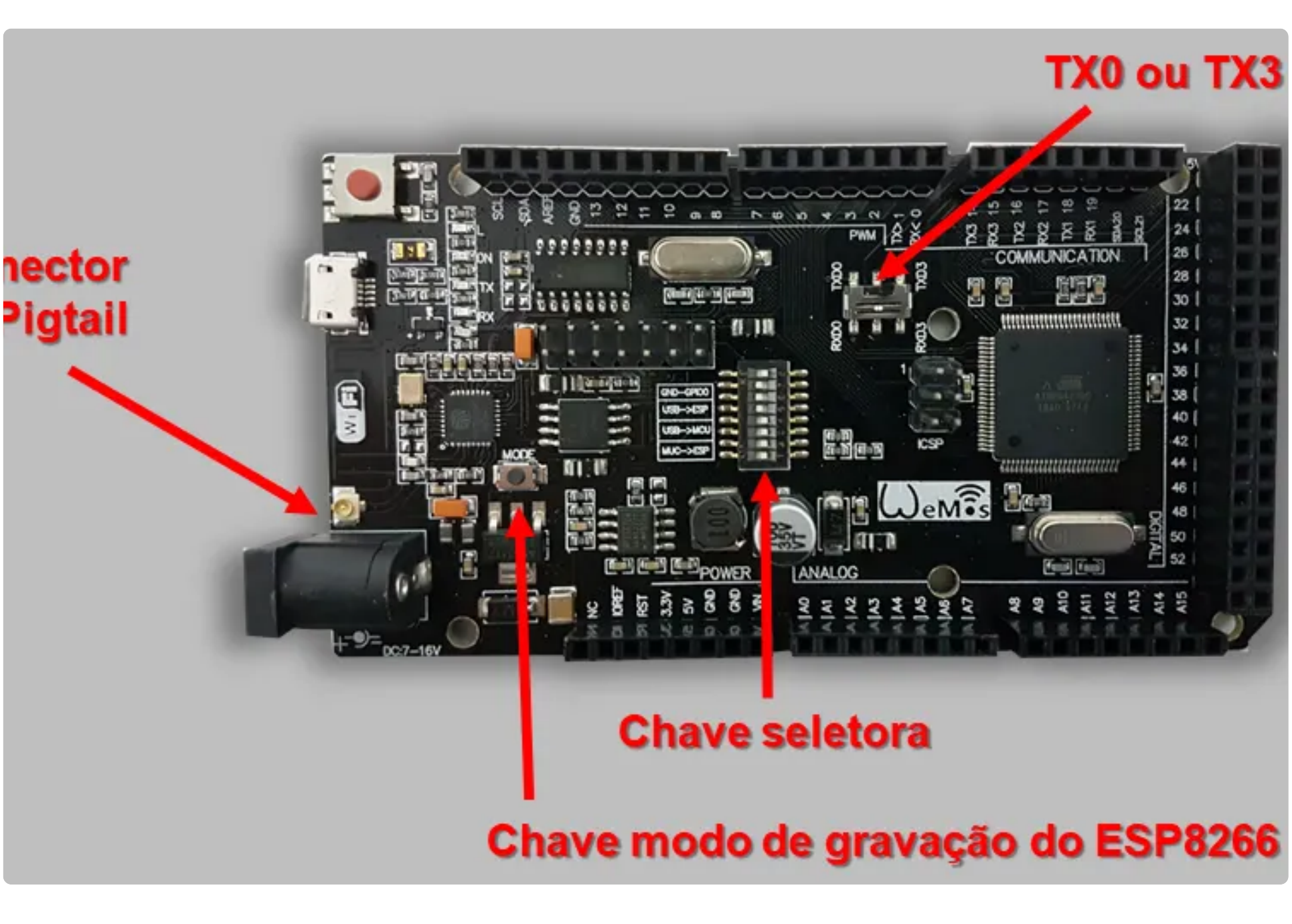
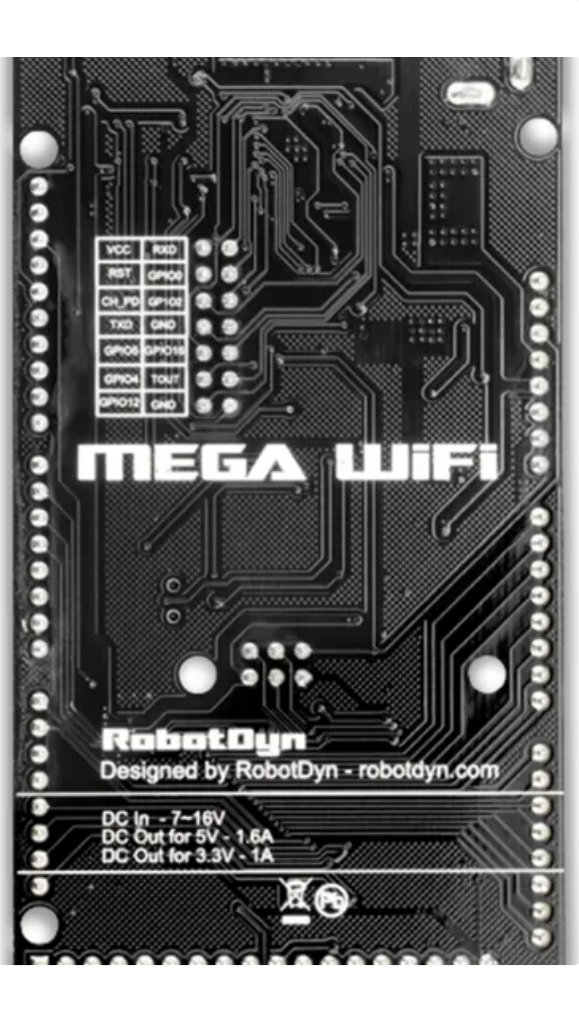
Step 1: Physical Characteristics of the Board

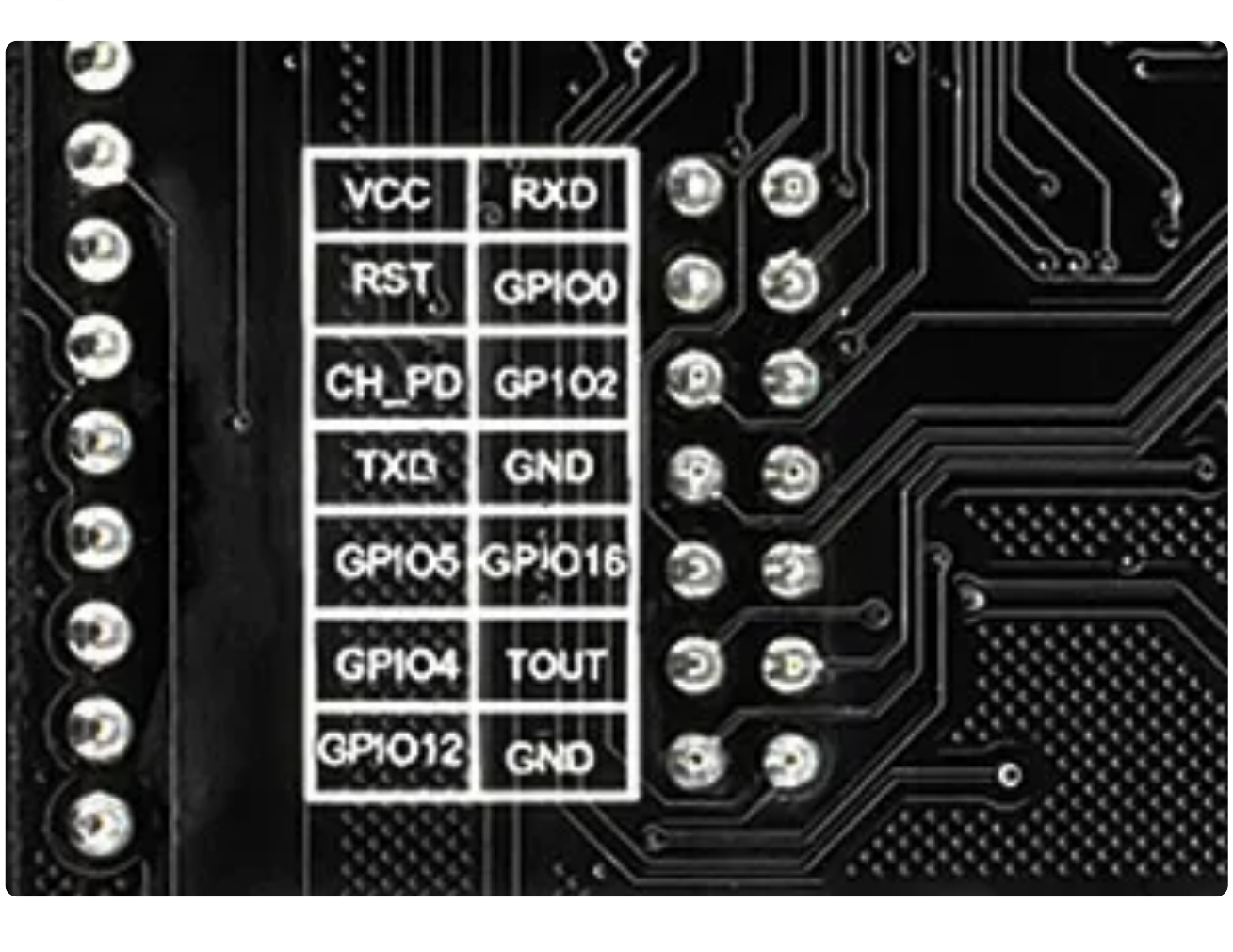

TX0 ou TX3
nector
Pigtail
Chave seletora
Chave modo de gravação do ESP8266

I really like that this Arduino has a Pigtail connector for an antenna. Why is this good? If you connect an antenna on this device, you will have a huge benefit, as it will increase your reach, straight from 90 meters to 240 meters away. I came to this conclusion after a test I did, so I did not have to rely solely on the datasheet manual.

This board also has a selector switch that allows the ESP to interleave the connection between TX0 and TX3, remembering that the ATmega has four serials. A second selector switch is the DIP Switch, and we also have a key recording mode of the ESP8266. All the pinning is completely compatible with the ATmega pinout.

Step 2: Access to ESP8266 Pins

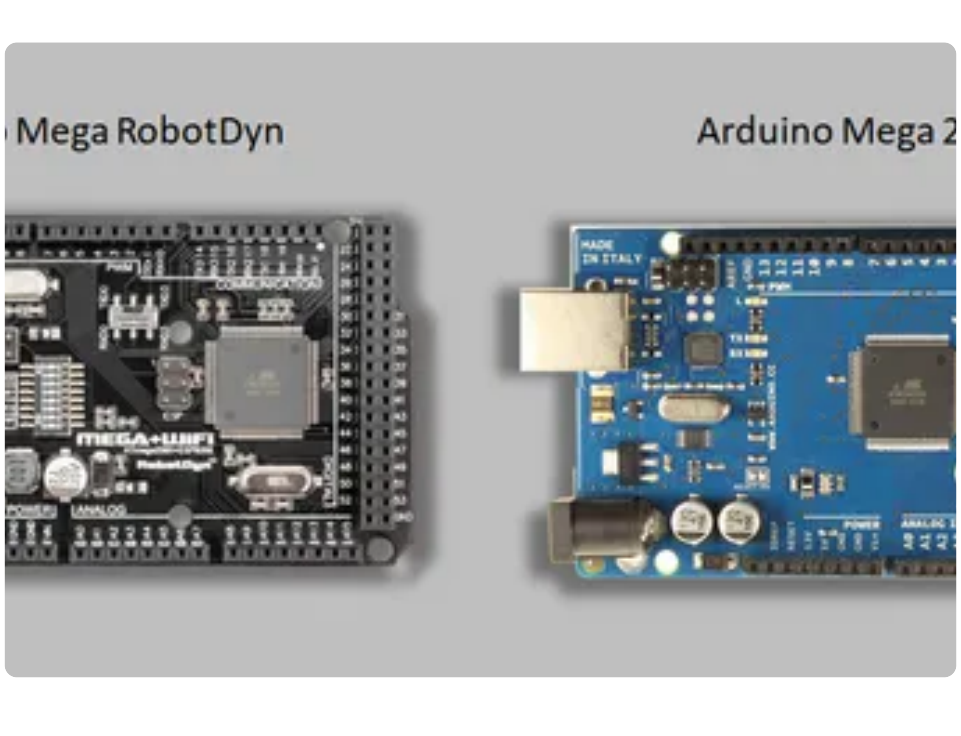

MEGA WiFi
RobotDyn
Designed by RobotDyn - robotdyn.com
DC in - 7~16V
DC Out for 5V - 1.6A
DC Out for 3.3V - 1A

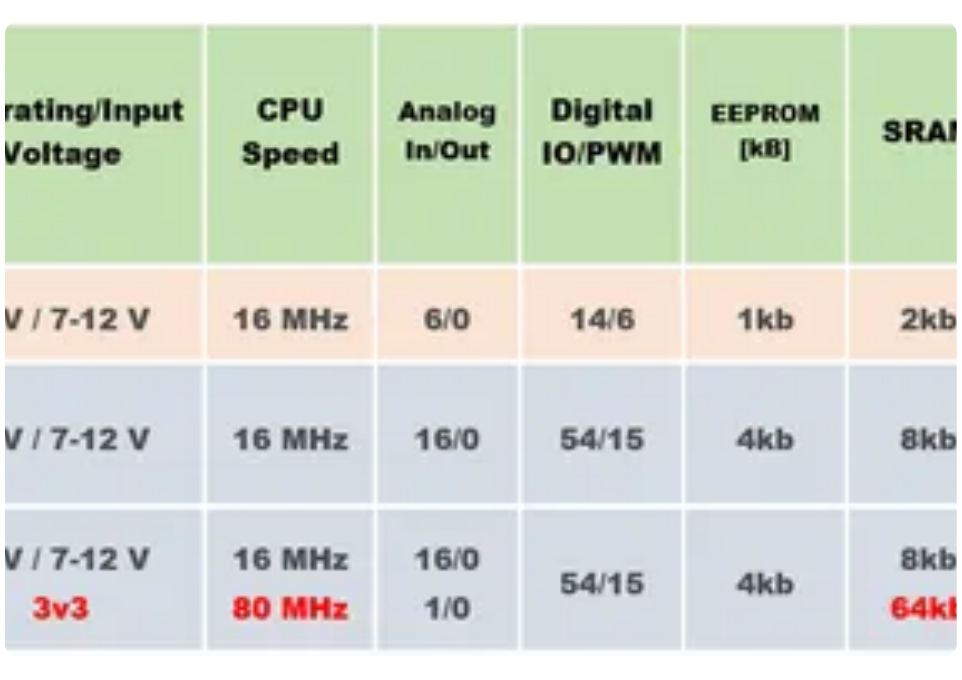


VCC	RXD
RST	GPIO0
CH_PD	GPIO2
TXD	GND
GPIO5	GPIO16
GPIO4	TOUT
GPIO12	GND

Here, I show the back of the board, where there is a table that displays access to the ESP pins.

Step 3: Comparing the Two Arduinos


Mega RobotDyn
Arduino Mega 2



rating/Input Voltage	CPU Speed	Analog In/Out	Digital IO/PWM	EEPROM [kB]	SRA
V / 7-12 V	16 MHz	6/0	14/6	1kb	2kb
V / 7-12 V	16 MHz	16/0	54/15	4kb	8kb
V / 7-12 V 3v3	16 MHz 80 MHz	16/0 1/0	54/15	4kb	8kb 64k

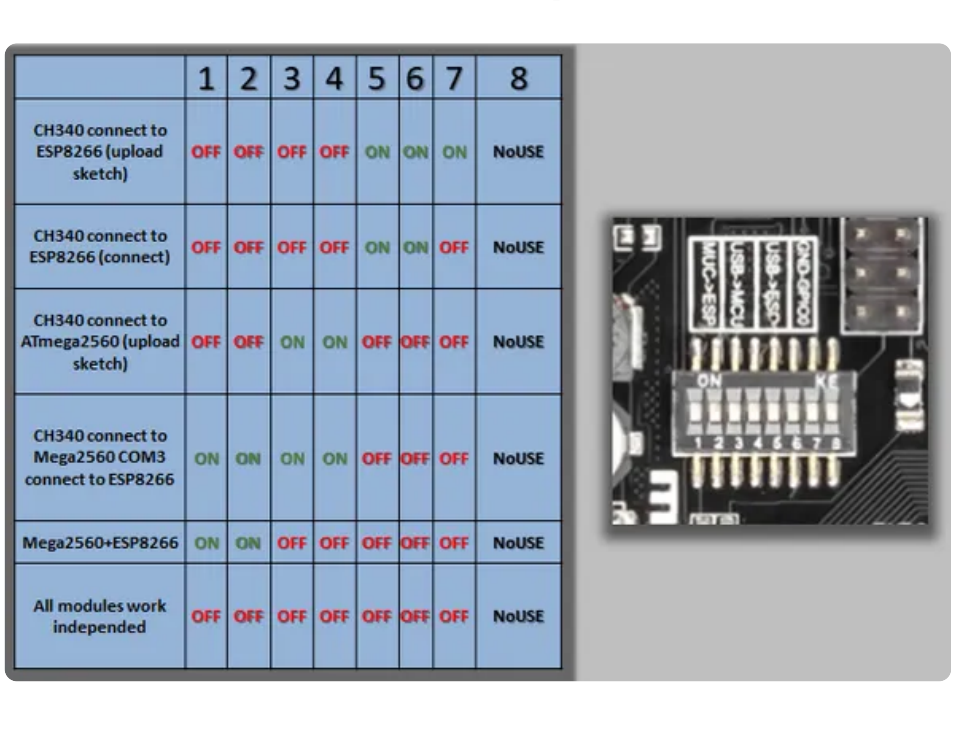
Here, we have a comparison between the Mega Arduino with the integrated ESP (Arduino Mega RobotDyn) and the traditional Mega Arduino (Arduino Mega 2560). We can see that they are similar, but in the 2560, we have the USB printer, which is a large connector. However, in the RobotDyn, we have the mini-USB. I particularly like the more compact option, but the power is the same in both.

We can see, then, that the intention of the RobotDyn creaters was to maintain the architecture of the ATmega.

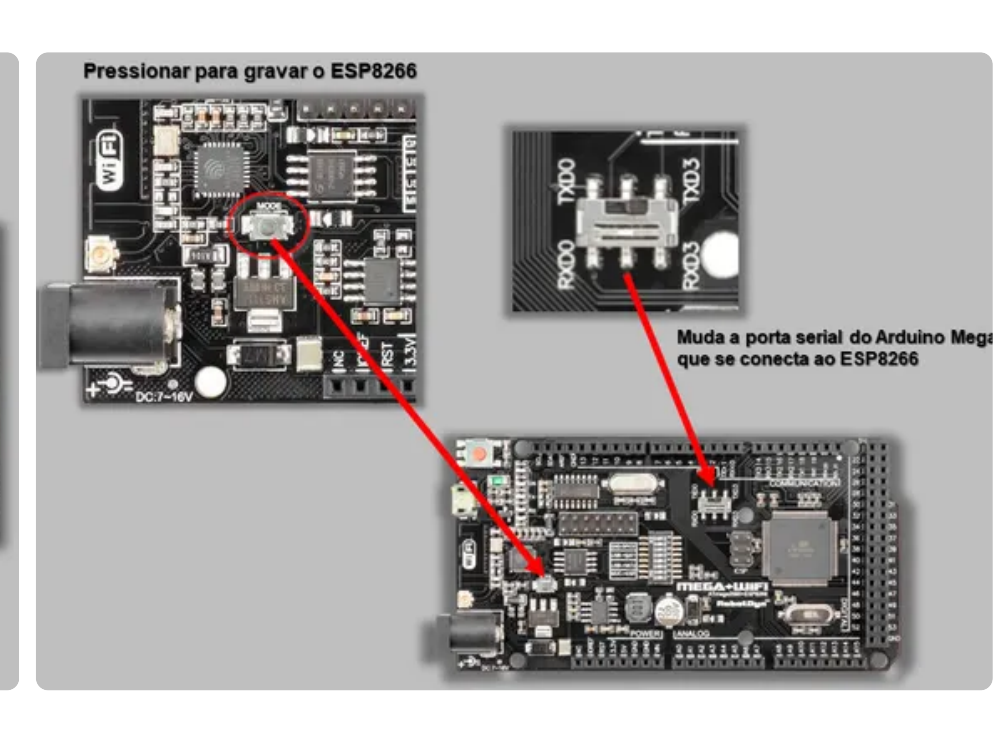
As we see in the table above, the ATmega has 32MB of memory, not counting the ESP memory. This is wonderful, since the traditional Mega Arduino has only 256kb of memory. The power in the RobotDyn is 7 to 12 volts, and the ESP8266 is already powered, and already has a voltage reducer. So, feeding the Arduino is already feeding the ESP, which is already down to 3v3, and internally that microcontroller is already 3v3.

The processors are the same, 16MHz, and one great advantage of these models is the high amount of IOs.

Step 4: Switch Status and Mode Selection



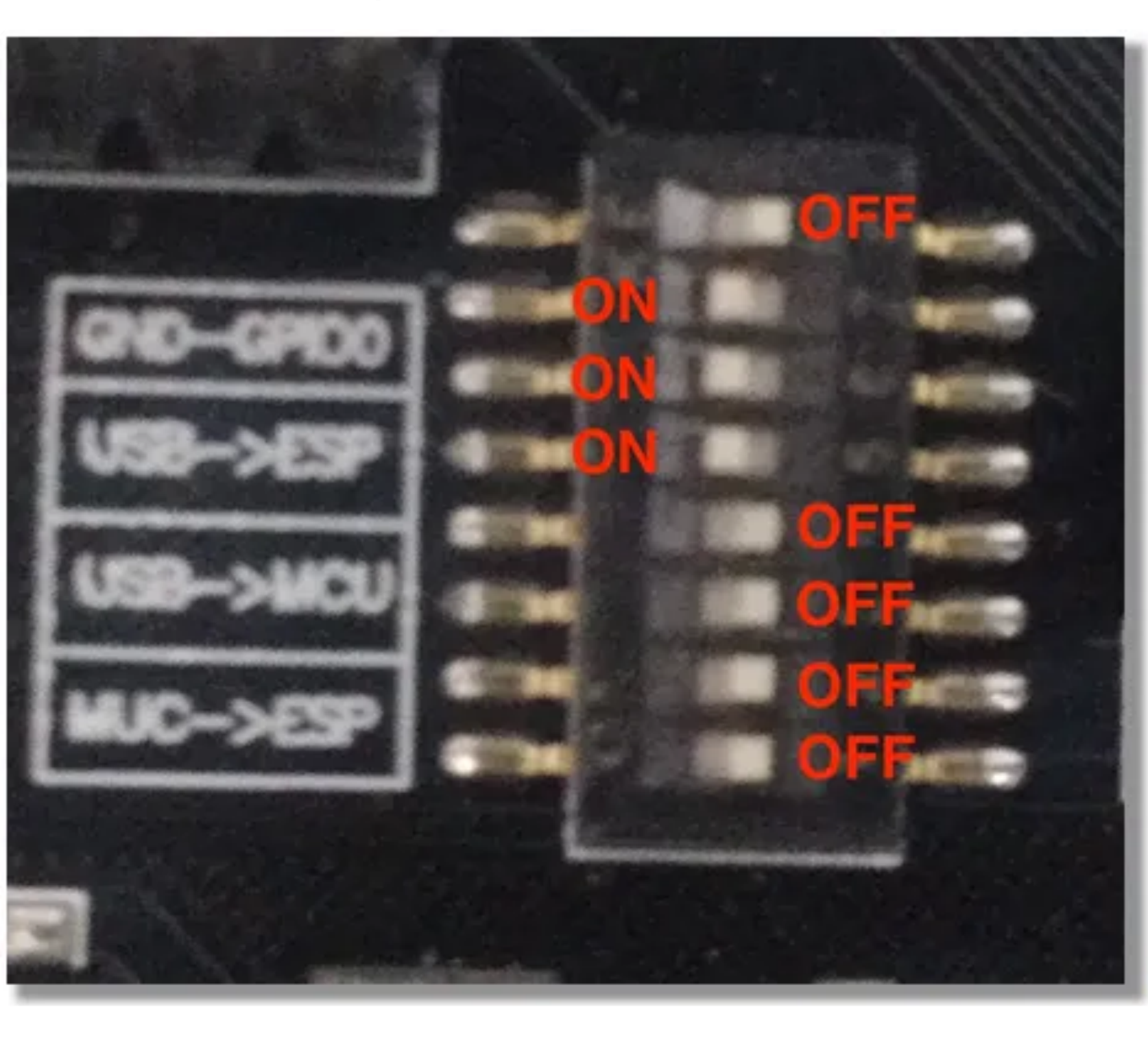
	1	2	3	4	5	6	7	8
40 connect to 8266 (upload sketch)	OFF	OFF	OFF	OFF	ON	ON	ON	NoUSE
40 connect to 266 (connect)	OFF	OFF	OFF	OFF	ON	ON	OFF	NoUSE
40 connect to ga2560 (upload sketch)	OFF	OFF	ON	ON	OFF	OFF	OFF	NoUSE
40 connect to ga2560 COM3 ect to ESP8266	ON	ON	ON	ON	OFF	OFF	OFF	NoUSE
2560+ESP8266	ON	ON	OFF	OFF	OFF	OFF	OFF	NoUSE
nodules work depended	OFF	OFF	OFF	OFF	OFF	OFF	OFF	NoUSE


Pressionar para gravar o ESP8266
Muda a porta serial do Arduino Mega que se conecta ao ESP8266

We have here the DIP Switch and a table with several positions. These help in the connections, depending on your objective. One essential detail is that if you are writing Flash in ESP, you have to be aware of addresses that are even slightly different.

In the image below, we zoomed in on the key that changes the serial port of the Arduino Mega. This connects to the ESP, and also in the key Mode, requiring that we must press the ESP8266 to record.

Step 5: AT Firmware Installation


GND-GPIO0
USB->ESP
USB->MCU
MUC->ESP
OFF
ON
ON
ON
OFF
OFF
OFF
OFF

If you want to use ESP8266 in AT mode, download the PDF file. You should now configure the card so that the ESP8266 is connected to the USB and in recording mode. To do this, set switches 5, 6, and 7 to ON (left) and all other switches to OFF (right).

If you want to use ESP8266 in AT mode, you should configure the Flash Download Tool as follows:

SPI Speed = 80MHz

SPI Mode = DIO

Flash Size = 32Mbit 4mb bytes x 8 bits = 32m bits

Crystal Freq = 26M

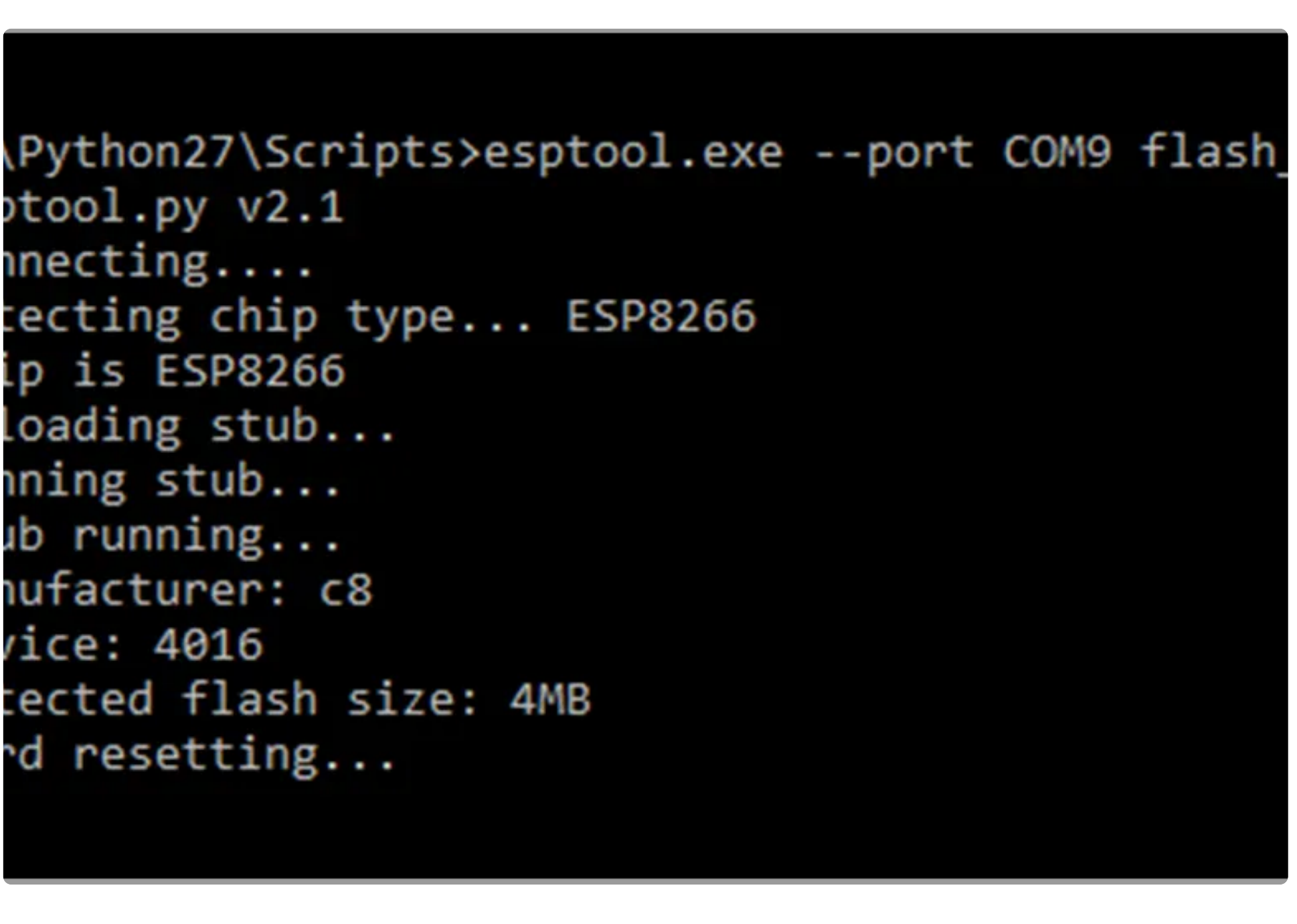
File\bin\esp_init_data_default.binataddress0x3fc000

File\bin\blank.binataddress0x37e000

File\bin\boot_v1.4(b1).binataddress0x00000

File\bin\at\512+512\user1.1024.new.2.binataddress0x1000

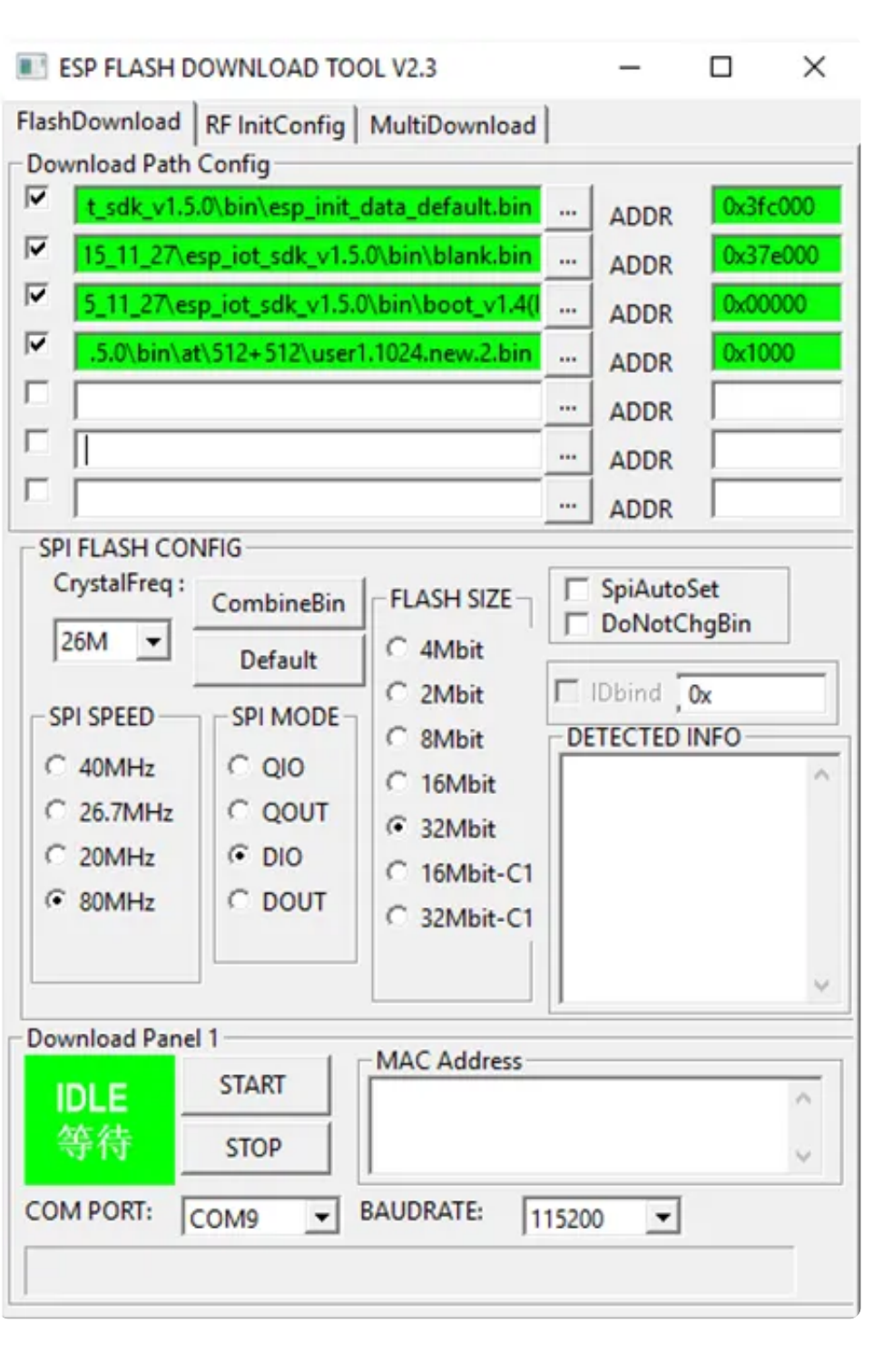
Step 6: Verifying AT Firmware



```
\Python27\Scripts>esptool.exe --port COM9 flash_
ptool.py v2.1
nnecting....
ecting chip type... ESP8266
ip is ESP8266
loading stub...
nning stub...
ub running...
nufacturer: c8
vice: 4016
ected flash size: 4MB
rd resetting...
```

In this part, I used esptool.exe, which is a command tool that accesses the ESP8266 Flash and checks for some settings, such as the chip type and memory size.

Step 7: Example


ESP FLASH DOWNLOAD TOOL V2.3
FlashDownload
RF InitConfig
MultiDownload
Download Path Config
SPI FLASH CONFIG
CombineBin
Default
SpiAutoSet
DoNotChgBin
IDbind
DETECTED INFO
Download Panel 1
IDLE
等待
START
STOP
MAC Address

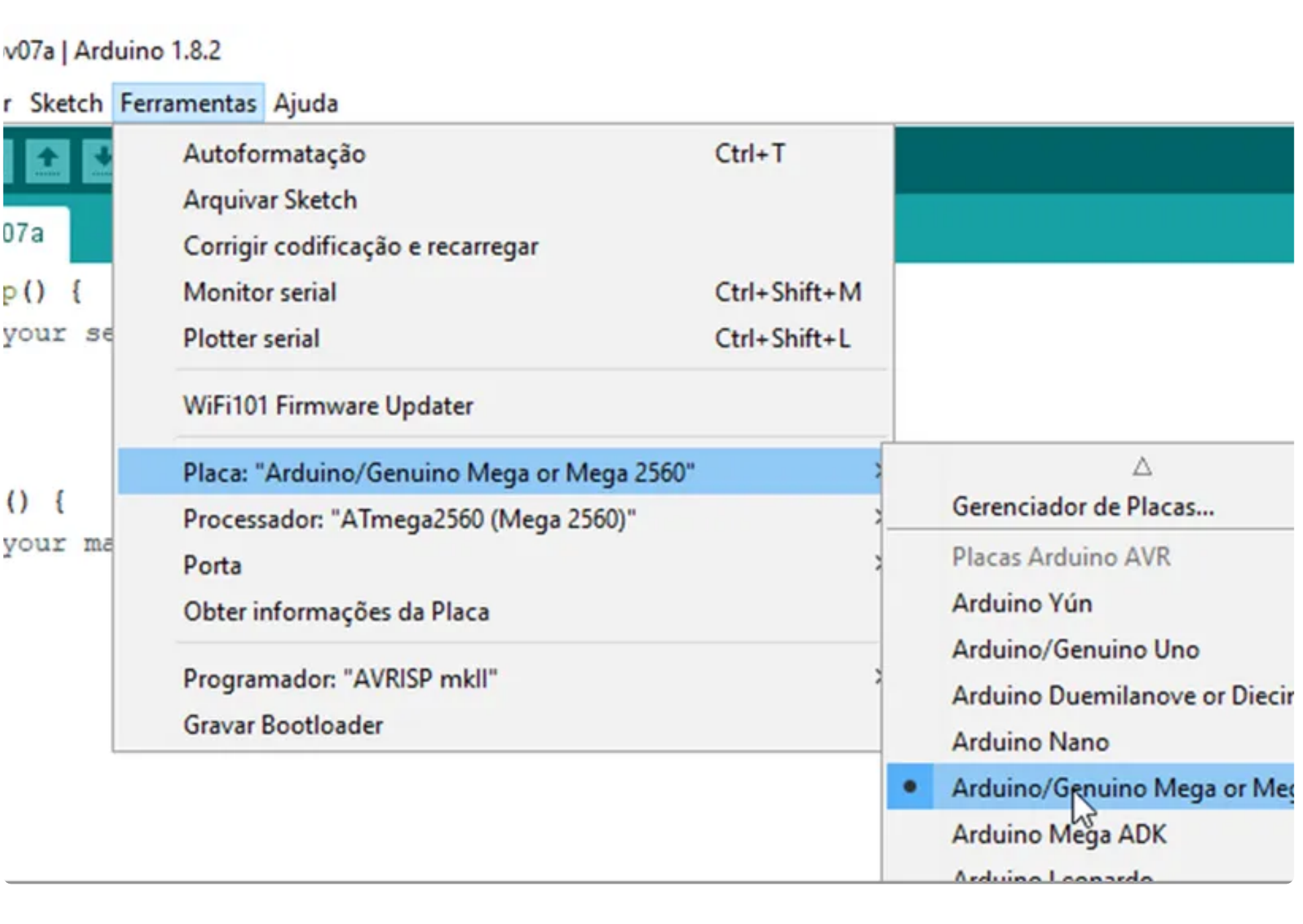
☑ t_sdk_v1.5.0\bin\esp_init_data_default.bin ADDR 0x3fc000
☑ 15_11_27\esp_iot_sdk_v1.5.0\bin\blank.bin ADDR 0x37e000
☑ 5_11_27\esp_iot_sdk_v1.5.0\bin\boot_v1.4(ADDR 0x00000
☑ .5.0\bin\at\512+512\user1.1024.new.2.bin ADDR 0x1000

CrystalFreq: 26M
SPI SPEED: ○ 40MHz ○ 26.7MHz ○ 20MHz ● 80MHz
SPI MODE: ○ QIO ○ QOUT ● DIO ○ DOUT
FLASH SIZE: ○ 4Mbit ○ 2Mbit ○ 8Mbit ○ 16Mbit ● 32Mbit ○ 16Mbit-C1 ○ 32Mbit-C1
COM PORT: COM9
BAUDRATE: 115200

In this example we show the hexadecimal addresses we use to write with the Flash Download Tool.

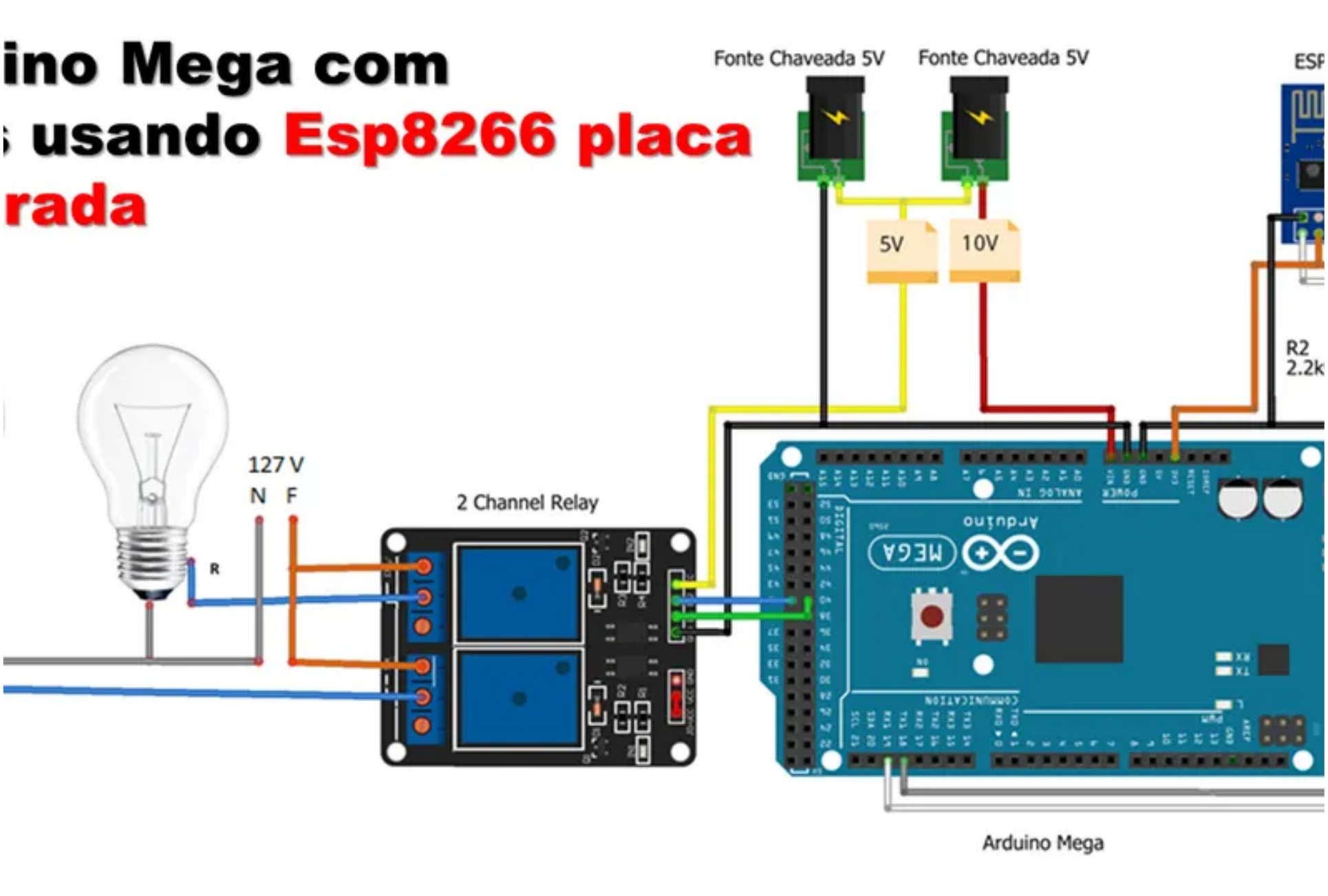
Also, for those who do not have much experience with the ESP8266, I suggest two of my previous videos: Recording in ESP01 and Introduction to ESP8266.

Step 8: Configure Arduino IDE Environment


v07a | Arduino 1.8.2
Sketch
Ferramentas
Ajuda
Autoformatação Ctrl+T
Arquivar Sketch
Corrigir codificação e recarregar
Monitor serial Ctrl+Shift+M
Plotter serial Ctrl+Shift+L
WiFi101 Firmware Updater
Placa: "Arduino/Genuino Mega or Mega 2560"
Processador: "ATmega2560 (Mega 2560)"
Porta
Obter informações da Placa
Programador: "AVRISP mkII"
Gravar Bootloader
Gerenciador de Placas...
Placas Arduino AVR
Arduino Yún
Arduino/Genuino Uno
Arduino Duemilanove or Diecir
Arduino Nano
Arduino/Genuino Mega or Meg
Arduino Mega ADK

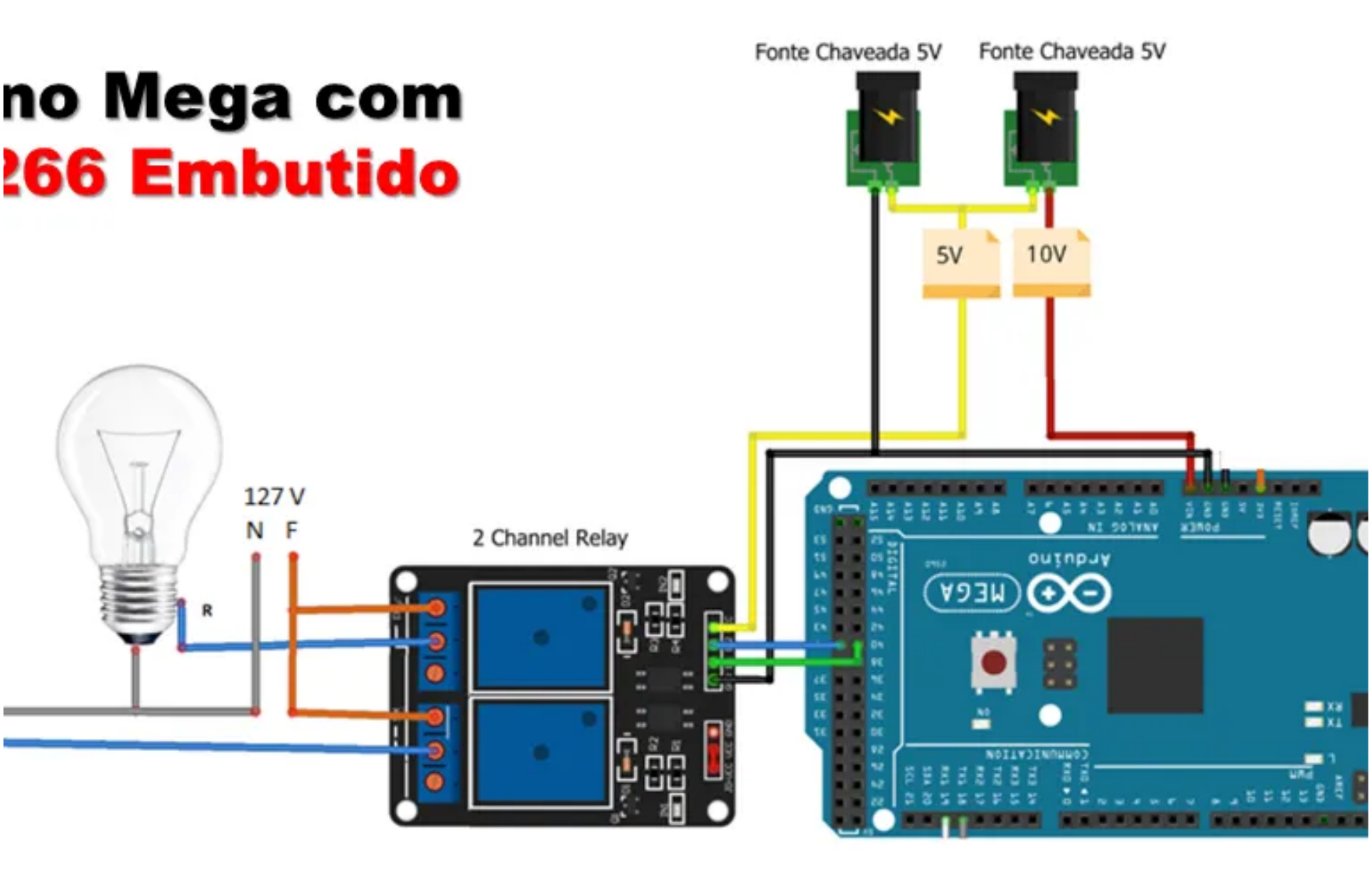
To record the Arduino, there is no mystery at all. You only need to set up the Mega Arduino 2560 board as if it were a traditional Arduino.

Step 9: Arduino Mega With Relays Using Esp8266 Separate Board


ino Mega com
usando Esp8266 placa
rada
Fonte Chaveada 5V
Fonte Chaveada 5V
ESP
5V
10V
R2
2.2k
127 V
N F
R
2 Channel Relay
Arduino Mega

We have here the assembly scheme that I perform in the video. We connected the Arduino Mega to the ESP01 and controlled two relays for one application.

Step 10: Arduino Mega With Built-in Esp8266


no Mega com
266 Embutido
Fonte Chaveada 5V
Fonte Chaveada 5V
5V
10V
127 V
N F
R
2 Channel Relay

Here, we do the same thing mentioned above, but while using the Arduino Mega with integrated ESP. One tip is that you watch the video entitled Residential Automation with Arduino Mega and ESP8266 for more details about this subject.

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Stone, Concrete, and Cement
Contest

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57 Comments

wakko59 a year ago

Hi, I just got one of these for a project for a Sauna control on Arduino Cloud. Is there a sketch that shows how i/o on the Mega is controlled via the ESP 8266 and an explanation of what dip switches are used to achieve this. The article above describes the control of 2 relays but is there a link to the sketch

Very new to this but I'm sure I can do it

benni7982 3 years ago

Hi, is there a
code available for changing the PWM frequencies of the selected pins
to stop DC motors from humming? For Arduino Uno I found something which worked
well but does not with this particular board... Would be great to get some help. :-) Cheers Ben

john henry 6 years ago

what the heck is "AT mode"? and why should or shouldn't i use it?
im not new to programming or electronics but im fairly new to arduino and esp development.
when i see "AT mode" first thing i think of is Auto Translate. which doesn't make sense.
i kinda hate acronyms especially when someone expects everyone to know what the Fun it means.

Acquired Target?
Aqueous Transition?
Abridged Translation?
Armored Transport?
American Tea?
Artistic Talent?
Ancient Transcripts?
Antichrist Transformation?
Almighty Tallest?
Acronym Translator?

▶ 14 replies

CuriousbotKC 3 years ago

Hello, can you please suggest what type of antenna should be attached? If you can provide any specs that will be great.

rodcoles 3 years ago

Hi Fernando,
It seems on most ESP32 boards, you have to remove or bypass a resistor to make the external antenna work. I have tried adding a pigtail and it makes no difference so I was wondering if you could explain what you have to do to alter the board for an external antenna to work?
Thank you!

contactather 4 years ago

Hi! Can I upgrade sketch of Mega, using ESP8266 OTA ?

Worka 4 years ago

My Board needs not only DIP Switches 3 and 4 but also DIP Switch 6 to be set ON for Uploading Sketch to Mega2560.

Some extra Information:
Mega2560 connects to the ESP8266 Serial1.
The Serial0 TX/RX Pins of the ESP8266 are some of the accessable ESP8266 Pins.

LorenzoS2 4 years ago

Hi, how can I do to communicate via serial from MEGA to ESP8266, with Serial3?

FranciscoR144 5 years ago

MUCHAS GRACIAS, Por la AYUDA., Eres muy amable gracias. CIUDAD DE MÉXICO., RAFAEL 🌈🐺

Bebel27 5 years ago

Hie, do you have the shetch example please ?

▶ 1 reply

Cool Zombie 5 years ago

Hi!
I m fairly new to Arduino environment,
I wanted to know that what should be the DIP switch and tx selector switch config. for uploading the code to atmega chip directly?

Also, is it possible to use this board without the wifi thingy for the time being?
I.e., use it as a usual mega board?

▶ 1 reply

gm310509 6 years ago

I would like to run the ESP8266 in AT mode, but I am completely confused about your statement to "download the PDF file" (step 5) - how does that help me run it in AT mode? Where can I get the AT mode firmware for the ESP8266 it doesn't seem to be in any of the samples. Maybe I am missing something, but I can not seem to find a reference anywhere on this page!?!?

▶ 1 reply

obiora 6 years ago

Good day all, please I need help here I have mega 2560+ Esp8266 I have tried to upload sketch and its not working but always giving me error massage

▶ 3 replies

Bulldog1989 6 years ago

How do I use the 'Flash Download Tool'; finally managed to download & open the program, enter the data you have given us, but nothing happens.

Also you have given a link to a youtube video, unfortunately it's not in English, so is no help to me.

Where do I find 'esptool.exe' ?

I have been working on getting this board to work for a week now, & starting to believe it is impossible to connect it to the internet.

Thanks

Alan

▶ 2 replies

isakitela 6 years ago

Where can I get the codes for this project? and once I have them .... Which code is for esp and which is for Mega?

l.dijkman 6 years ago

Would like to make a thermostat
I am lost, need a reset ;-)
My wifi board (ordered similar mega)
https://youtu.be/3hYfkGEpPko
My thermostat
https://youtu.be/lny7J7KXk9w

mrsharkit 8 years ago

hi
to enable external antenna, isn't it required to remove or move one of the 2 resistors near the socket? I read it somewhere, but can't find info on it...

▶ 3 replies

Mauricio Holguin 6 years ago

Hello everyone, I have a WEMOS D1 R3 board, and so far I have access to the ESP8266 mode it was connected to wifi (PIN 0,0,0,0,1,1,1, -), but to access all the pins of the card separates the codes, between ESP8266 and MEGA 2560 (PIN 1,1,1,1,0,0,0, -). Although the wemos permance card is connected to the wifi, on the MEGA 2560 side, I cannot connect to MySql, it seems to be the card in these two modules are each independent. I would like you to help me solve this problem of how to connect to MySql with WEMOS D1 R3 MEGA by MEGA 2560 mode (PIN 1,1,1,1,0,0,0, -) since I need to use a lot of pins for the project I'm developing. Thank you very much for your help.

JonM189 6 years ago

Hi all,

has anyone used this linking the SCA SCL pins on the mega to the gpio pins on the esp8266?

I was hoping to use esphome with the mega as an i2c port expander? if so, would they mind sharing how it worked.

I am assuming i would just be able to use jumper wires to connect, but would there be a 5v 3.3v issue?

Many thanks in advance

justtolearnsfu 7 years ago

Hy. My idea was to use the esp8266 with AT firmware, create a simple webserver, and create or copy some code with the neopixel library to control a led strip. First of all, this esp is weird. I've tried a lot of combinations to upload a firmware to it, but I have no answer, or a simple "ERROR" message. The way I am doing it /most of the time/ is dip switch 5 6 7 ON, connect usb, reset, using the values from your instructions, push "mode" button, start uploading, everything looks good. Then disconnect, dip switch 5 6 ON, connect, and in serial monitor I have nothing or error, depending on the version I've uploaded. Is there a magic combo I haven't found yet? Please somebody help me out, once I had given up, but I have nothing else to start with, so I am trying combinations again 12 hours a day in the last 3 days, and honestly, it freaks me out. On to of it, sometimes I get this short message at 74880 baud, but the ERROR message for the AT command comes at 115200. I hope you can help me because if it will be solved, it could be a hell of a story, otherwise it will be a great disappointment.
Thanks in advance!

dansum 7 years ago

I can upload code to the 2560 and the esp8266 with no problem thanks to your help. I can receive serial port output in the serial monitor from the 2560 but can not get any output when programming the esp8266. I have tried moving the dip switches to different positions but have not been able to get any output at all. I have searched but haven't found much about this Mega 2560 with the esp8266 built on it. I am using board NodeMCU 1.0(esp12E) and am using com 6. It uploads great but no serial output at all. Do you have any suggestions?

▶ 1 reply

Neuroplant 7 years ago

Hello,
you are writing about "the Flash Download Tool" in Chapter "Step 5: AT Firmware Installation" where di I get it? Does it have a Name?

Neuroplant

cemal kaplan 7 years ago

Problem Ok, Thank you very much...

minassianhagop 7 years ago

Hello, I want to use the board as arduino mega without ESP2866. and I can't upload the code, even though I set the switches as 0011000 USB <->ATmega2560 is there anything else should I do ?

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